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EDUCATION FOR ENTREPRENEURSHIP IN HIGHER EDUCATION: ACTIVE METHODOLOGIES, ESG AND CHALLENGES OF DIGITAL TRANSFORMATION

Jardiel de Moura Gomes[1]
Fernando Gomes de Paiva Junior[2]
Riquelme Batista Gonçalo[3]
Victor Gabriel Corrêa do Nascimento Feitoza[4]
Marcela Araújo de Lima[5]

ABSTRACT

Objective: This study analyzes how active methodologies, with an emphasis on Problem-Based Learning (PBL) and gamification, contribute to Entrepreneurship Education in higher education by incorporating ESG perspectives and addressing the challenges posed by digital transformation, based on the perceptions of students and faculty from a private institution.

Theoretical Framework: The research is grounded in references on Entrepreneurship Education, active methodologies, ESG, and pedagogical innovation, highlighting the role of the educational entrepreneur as a mediator of the integration between teaching practices, digital transformation, and sustainable values.

Method: A mixed and descriptive approach was adopted. Questionnaires were applied to 66 Business Administration students, between June 2 and 27, 2025, and three semi-structured interviews were conducted with specialists. Quantitative data were analyzed using descriptive statistics, and qualitative data were examined through content analysis.

Results and Discussion: The findings indicate that PBL and gamification enhance student motivation, engagement, and protagonism, fostering critical, creative, and collaborative skills. The integration of ESG principles proved relevant to entrepreneurial education, although limited by institutional and structural barriers and by the need for better teacher preparation in the context of digital transformation.

Implications and Originality: The study reinforces the importance of pedagogical intentionality and the role of the educational entrepreneur in articulating active methodologies, ESG, and technological innovation. The originality lies in highlighting the convergence of these elements in Entrepreneurship Education, emphasizing the strategic role of the educational entrepreneur in building an innovative, ethical, and socially committed university.

Keywords: Entrepreneurship Education, Active Methodologies, ESG, Digital Transformation, Higher Education.

EDUCAÇÃO PARA O EMPREENDEDORISMO NO ENSINO SUPERIOR: METODOLOGIAS ATIVAS, ESG E DESAFIOS DA TRANSFORMAÇÃO DIGITAL

RESUMO

Objetivo: Este estudo analisa como metodologias ativas, com ênfase na Aprendizagem Baseada em Problemas (ABP) e na gamificação, contribuem para a Educação para o Empreendedorismo no ensino superior, incorporando perspectivas ESG e enfrentando os desafios impostos pela transformação digital, a partir da percepção de estudantes e docentes de uma instituição privada.

[1] Centro Universitário Estácio do Recife, Recife, Pernambuco, Brazil. E-mail: jardielmmoura@gmail.com
[2] Universidade Federal de Pernambuco (UFPE), Recife, Pernambuco, Brazil. E-mail: fernando.paivajr@ufpe.br
[3] Centro Universitário Estácio do Recife, Recife, Pernambuco, Brazil. E-mail: riquelmebatista2014@gmail.com
[4] Centro Universitário Estácio do Recife, Recife, Pernambuco, Brazil. E-mail: victor.g.correa25@gmail.com
[5] Centro Universitário Estácio do Recife, Recife, Pernambuco, Brazil. E-mail: marcela.sula@gmail.com

Referencial Teórico: A pesquisa fundamenta-se em referenciais sobre Educação para o Empreendedorismo, metodologias ativas, ESG e inovação pedagógica, destacando o papel do empreendedor educacional como mediador da integração entre práticas pedagógicas, transformação digital e valores sustentáveis.

Método: Adotou-se abordagem mista e descritiva. Foram aplicados questionários a 66 estudantes de Administração, entre 2 e 27 de junho de 2025, e realizadas três entrevistas semiestruturadas com especialistas. Os dados quantitativos foram tratados por estatística descritiva e os qualitativos por análise de conteúdo.

Resultados e Discussão: Os achados indicam que ABP e gamificação favorecem motivação, engajamento e protagonismo discente, promovendo competências críticas, criativas e colaborativas. A integração de princípios ESG mostrou-se relevante para a formação empreendedora, embora limitada por barreiras institucionais, estruturais e pela necessidade de maior preparo docente diante da transformação digital.

Implicações e Originalidade: O estudo reforça a importância da intencionalidade pedagógica e da atuação do empreendedor educacional na articulação entre metodologias ativas, ESG e inovação tecnológica. A originalidade reside em evidenciar a convergência entre esses elementos na Educação para o Empreendedorismo, ressaltando o papel estratégico do empreendedor educacional na construção de uma universidade inovadora, ética e socialmente comprometida.

Palavras-chave: Educação Empreendedora, Metodologias Ativas, ESG, Transformação Digital, Ensino Superior.

EDUCACIÓN PARA EL EMPRENDIMIENTO EN LA EDUCACIÓN SUPERIOR: METODOLOGÍAS ACTIVAS, ESG Y RETOS DE LA TRANSFORMACIÓN DIGITAL

RESUMEN

Objetivo: Este estudio analiza cómo las metodologías activas, con énfasis en el Aprendizaje Basado en Problemas (ABP) y la gamificación, contribuyen a la Educación para el Emprendimiento en la educación superior, incorporando perspectivas ESG y afrontando los desafíos impuestos por la transformación digital, a partir de la percepción de estudiantes y docentes de una institución privada.

Marco Teórico: La investigación se fundamenta en referentes sobre Educación para el Emprendimiento, metodologías activas, ESG e innovación pedagógica, destacando el papel del emprendedor educativo como mediador de la integración entre prácticas de enseñanza, transformación digital y valores sostenibles.

Método: Se adoptó un enfoque mixto y descriptivo. Se aplicaron cuestionarios a 66 estudiantes de Administración, entre el 2 y el 27 de junio de 2025, y se realizaron tres entrevistas semiestructuradas con especialistas. Los datos cuantitativos fueron analizados mediante estadística descriptiva y los cualitativos a través de análisis de contenido.

Resultados y Discusión: Los hallazgos indican que el ABP y la gamificación favorecen la motivación, el compromiso y el protagonismo estudiantil, promoviendo competencias críticas, creativas y colaborativas. La integración de los principios ESG resultó relevante para la formación emprendedora, aunque limitada por barreras institucionales y estructurales y por la necesidad de una mejor preparación docente frente a la transformación digital.

Implicaciones y Originalidad: El estudio refuerza la importancia de la intencionalidad pedagógica y del papel del emprendedor educativo en la articulación de metodologías activas, ESG e innovación tecnológica. La originalidad radica en evidenciar la convergencia de estos elementos en la Educación para el Emprendimiento, subrayando el papel estratégico del emprendedor educativo en la construcción de una universidad innovadora, ética y socialmente comprometida.

Palabras clave: Educación para el Emprendimiento, Metodologías Activas, ESG, Transformación Digital, Educación Superior.

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1 INTRODUCTION

Digital transformation, combined with demands for more sustainable and inclusive practices, repositions the role of educational institutions as protagonists of a new training agenda. Entrepreneurship education (EfE) emerges as a strategic response to this scenario, by seeking to develop skills that articulate creativity, autonomy, innovation and social responsibility (Nabi *et al*., 2017; Guerrero; Urbano, 2012).

The insertion of EpE refers to a proposal that demands an epistemological and pedagogical repositioning with the student at the center of the formative experience. For this, the adoption of active methodologies stimulates protagonism, problem solving and meaningful learning. The scholar Paulo Freire highlights that learning only becomes significant when linked to the student's reality, their culture and their experiences, promoting transformation and social emancipation (Freire, 2014).

Among the active methodologies used in EpE in higher education, the flipped classroom, project-based learning, case studies, the use of *role playing* and simulations, and *Design Thinking* stand out. Among these methodologies, Problem-Based Learning (PBL), focused on the investigation of complex and open problems, and gamification, which integrates game elements to make the educational process engaging and motivating, which contribute to the development of critical, collaborative and entrepreneurial skills (Din *et al*., 2016; Hardie *et al*., 2020).

By confronting students with real or simulated problems, PBL promotes the articulation between theory and practice, stimulating autonomy, research and teamwork, since it breaks with the traditional instructional logic and values the process of collective construction of knowledge. Already gamification (*gamification*), by incorporating game elements such as challenges, rewards and narratives, enhances intrinsic motivation and creates dynamic and engaging training environments (Din *et al*., 2016; Hardie *et al*., 2020).

The integration of ESG (*Environmental, Social and Governance*) principles to entrepreneurship education extends the scope of training by including ethical values, environmental responsibility and governance practices as central dimensions of the educational process. This approach contributes to the construction of a critical awareness regarding the role of the entrepreneur in the face of social and *environmental challenges, a dimension still insufficiently discussed in studies on entrepreneurship education in higher education (Guerrero et al*., 2020).

The educational entrepreneur (*edupreneur*) emerges as a central pedagogical mediator. Unlike the traditional teacher, he acts at the interface between innovation, curriculum and institutional culture, promoting experiences aligned with digital transformations and social demands. Its actions include curating technologies, applying active methodologies and articulating with the innovation ecosystem, aiming to build an entrepreneurial and responsive university (Cui; Bell, 2022; Passarelli; Bongiorno, 2025).

This study dialogs with previous research (Moura Gomes *et al*., 2025), which analyzed the integration between active methodologies and entrepreneurship education in the context of digital transformation. However, it advances by explicitly incorporating ESG principles as an essential formative dimension and by problematizing the challenges arising from digital transformation for pedagogical practice. In addition, it is distinguished by presenting new empirical data, obtained through the application of 66 questionnaires to business students and three semi-structured interviews with experts, which allows a more comprehensive and in-depth analysis of the potential and limitations of adopting active methodologies in entrepreneurial training.

Given this scenario, **this study analyzes how active methodologies, with an emphasis on Problem Based Learning (PBL) and gamification, contribute to Education for Entrepreneurship in higher education, incorporating ESG perspectives and facing the challenges posed by digital transformation, from the perception of students and teachers.**

1.1 ACTIVE METHODOLOGIES IN ENTREPRENEURIAL EDUCATION

The adoption of active methodologies in higher education modernizes pedagogy, surpassing the traditional model of content transmission. In Entrepreneurship Education, these approaches create dynamic and collaborative environments, developing creativity and critical thinking. Therefore, active methodologies are based on constructivist approaches to learning through experimentation, reflection and interaction with real-world problems (Bacich; Moran, 2017; Costa *et al*., 2021). Such mobilization is essential to EoE, as it contributes to the internalization of entrepreneurial values and brings the educational process closer to the demands of the labor market and innovation ecosystems.

Active methodologies articulate theory and practice, overcoming the dichotomy between academic knowledge and concrete experience. They enable students to experience real situations of decision making, resource management, solution creation and risk assessment, in controlled but challenging contexts (Costa *et al*., 2021; Bortolaso *et al*., 2024). Through

immersive experiences, students are encouraged to take risks, experiment with solutions, and address the complexity of current social and economic challenges (Moura Gomes *et al*., 2025).

In the context of higher education institutions, the adoption of active methodologies requires structural, curricular and pedagogical reconfigurations (Araujo *et al*., 2024). Thus, the relevance of active methodologies for EoE is associated with the role of the educational entrepreneur as an agent of pedagogical innovation. By implementing active practices, the figure of the entrepreneurial educator - *edupreneur* promotes the renewal of the teaching-learning environment and prepares students to respond to the challenges of an economy based on knowledge and innovation (Malta *et al*., 2025).

1.2 PROBLEM-BASED LEARNING (PBL)

Abstract: Problem-Based Learning (PBL) is one of the active methodologies recognized in the educational field, especially in higher education, for its ability to promote autonomy, critical thinking and the resolution of complex problems. Unlike traditional teaching, centered on the exhibition of contents, PBL starts from the presentation of a real or simulated problem so that students collectively seek solutions, activating previous knowledge and developing new skills (Costa *et al*., 2021).

In Entrepreneurship Education, PBL simulates business challenges, stimulating creativity, critical analysis and decision making in contexts of uncertainty (Bortolaso *et al*., 2024; Malta *et al*.; 2025). Its effectiveness depends on the pedagogical intentionality and the teacher's role as mediator, which requires a review of traditional practices and teacher training focused on listening, reflective planning and collaboration (Børte *et al.;* 2023; Araujo *et al*., 2024). In the context of EfE, the teacher is responsible for structuring formative experiences focused on action and creative problem solving. The problems used in PBL must be authentic, contextualized and *challenging, in order to stimulate the intellectual and emotional engagement of students.* Therefore, the more the problem-situation is connected to reality, the propensity of students to develop transferable skills to professional contexts, including the ability to network, negotiate solutions and lead processes (Costa *et al*., 2021).

PBL favors collaborative learning, as it requires the articulation between different knowledge, perspectives and skills of the group members. By working as a team, students learn to deal with differences, to develop empathy and to share responsibilities, essential aspects in the exercise of leadership and in acting in entrepreneurial environments. This type of dynamics challenges the logic of individual competition present in higher education and promotes a

culture of cooperation and collective intelligence (Bortolaso *et al*., 2024; Moura Gomes *et al*., 2025). However, the implementation of PBL still finds resistance in higher education courses, due to the lack of adequate teacher training, or due to the structural limitations of the institutions. In regulated and bureaucratic contexts, the flexibility required for PBL is hampered.

To overcome these barriers, an institutional commitment to pedagogical innovation involving supportive policies and an academic *ethos* oriented to experimentation and student protagonism is necessary (Børte *et al.;* 2023; Malta *et al*., 2025).

Problem-Based Learning provides contextualized educational experiences aligned with real challenges, in addition to supporting the development of students' perceived self-efficacy, increasing their ability to engage in the face of multidimensional social and economic problems. By favoring an active, critical and situated learning, PBL consolidates itself as a fundamental formative strategy in Education for Entrepreneurship, promoting the constitution of autonomous, creative and oriented subjects for social transformation.

1.3 GAMIFICATION (*GAMIFICATION*)

Gamification (*Gamification*) has been consolidated as a relevant pedagogical strategy for EoE for its ability to mobilize engagement, motivation and development of entrepreneurial skills in formal teaching environments. This approach incorporates characteristic elements of games such as challenges, rewards, *rankings*, narratives and immediate *feedback* in educational activities, with *the objective of making the learning process engaging, participatory and effective (Moura Gomes et al*.; 2019; Ciprian, 2021).

By promoting autonomy and dynamic contexts, gamification strengthens entrepreneurial intention and the development of critical skills (Werbach & Hunter, 2012; Ciprian, 2021). In EpE, it proved to be effective in simulating decisions, resource management and problem solving, favoring experiential learning and the practical internalization of concepts.

This characteristic is evident in contexts marked by the digital transition when the ability to adapt quickly and creative problem solving become central skills for the formation of entrepreneurial subjects (Sailer *et al*., 2017). Gamification influences students' motivation by promoting a sense of progress, purpose and belonging in the learning environment (Deci & Ryan, 2013; Grivokostopoulou *et al*., 2019). However, it is necessary to recognize that the adoption of gamification requires pedagogical planning (Burak *et al*.; 2025). Thus, there is also

the didactic intentionality in gamification, so that the game elements are integrated into the learning logic (Ciprian, 2021).

Therefore, gamification promotes content retention, accelerates the development of socio-emotional skills and favors the protagonism of students in their learning journey, stimulates the desire to undertake, strengthens the ability of students to deal with risks, uncertainties and pressures of the entrepreneurial environment (Grivokostopoulou *et al*., 2019).

In educational environments marked by passivity and detachment, gamification reconfigures the role of the learner as an active agent of the teaching-learning process (Sailer *et al*., 2017; Werbach & Hunter, 2012; Burak *et al*.; 2025). The *edupreneur*, in turn, is responsible for adapting technological resources to the specificities of the curriculum and the institutional mission of entrepreneurial training. The performance of this agent requires not only technical mastery, but also political and epistemological sensitivity in the face of tensions that cross educational projects (Sailer *et al*., 2017; Grivokostopoulou *et al*., 2019; Burak *et al*.; 2025).

2 THEORETICAL FRAMEWORK

This section discusses Education for Entrepreneurship (EoE) in higher education, focusing on its pedagogical approaches, the challenges of implementation in HEIs and the role of the educational entrepreneur as an agent of innovation.

2.1 EDUCATION FOR ENTREPRENEURSHIP AND EDUCATIONAL ENTREPRENEURS

Abstract: Entrepreneurship Education (EfE) has gained prominence in the global educational agenda since the 1980s, driven by the recognition of the strategic role of small and medium-sized enterprises (SMEs) as alternatives to unemployment and the economic crisis in different countries (Pepin, 2018; UNESCO, 2020, 2024). In this context, it has become necessary to reformulate educational models in order to foster in students the ability to transform talents and skills into concrete, sustainable and innovative solutions.

EpE has promoted learning experiences oriented to the development of entrepreneurial skills, such as adaptability, creativity, collaboration and resilience (Lackéus, 2015; Hamdan, 2020). Some countries have been adopting systematic approaches to curricular integration at different educational levels. In Finland and Sweden, for example, PE is present from basic education to higher education, structuring the training path based on active methodologies and action orientation (Fejes *et al*., 2019).

In the United States, the term "entrepreneurial education" highlights the practice as a professional axis, while in the United Kingdom the terminology "business education" predominates, more associated with business structuring (Lackéus, 2015; Araujo *et al*.; 2024). In New Zealand, the integration of entrepreneurship occurs since the early years of formal education, with the aim of developing an innovative mindset from an early age. Conceptually, EpE can be understood from two main approaches: (i) narrow approach, centered on technical-business contents, such as *marketing*, finance and management; and (ii) broad approach, guided by interdisciplinary methodologies and project and problem based learning (Kirkley, 2017; Hardie *et al*., 2020). Both coexist in a complementary way by articulating theory, practice and real learning experiences (Lackéus, 2015; Fejes *et al*., 2019).

The simple addition of EoE to the curriculum does not guarantee transformations in academic education, because its success depends on authentic pedagogical experiences, which challenge students *to solve complex problems and develop socially relevant solutions.* In addition to the methods, recent studies highlight the role of instructors, the content and the profile of the target audience (Nabi *et al*., 2017), maintaining the PE programs linked to the university environment (Kirkwood *et al*., 2014; Passarelli & Bongiorno, 2025). Abstract: Institutions (HEIs) operate in the training of students for the labor market through approaches such as Education for Entrepreneurship (EoE). This stimulus contributes to its insertion in innovative initiatives and directed to the development of new businesses (Cui & Bell, 2022).

The relationship between EoE and entrepreneurial intention (EI) presents a dynamic and multidimensional character, requiring approaches that contemplate psychological and contextual aspects in the construction of the entrepreneurial profile. Entrepreneurial intention (EI) can be defined as the conscious willingness of an individual to start an enterprise at a future time (Liñán *et al*., 2011; Passarelli & Bongiorno, 2025). Abstract: The educational entrepreneur - *edupreneur* - is inserted as a strategic subject within universities, acting in the mediation between these institutional transformations and the concrete pedagogical practices. By proposing and implementing innovations in teaching, learning and academic management, it contributes to the consolidation of an adaptable, creative university connected to the demands of society and the market.

2.2 PEDAGOGY OF EDUCATION FOR ENTREPRENEURSHIP

Education plays a central role in the training of students, enabling not only theoretical knowledge, but also essential skills to face the challenges of an uncertain and dynamic global

scenario. Entrepreneurial training, in particular, offers a curriculum and essential skills that stimulate resilience, autonomy, innovation and the ability to identify opportunities, preparing students for a productive professional life (Ioannou & Retalis, 2025).

The study "Entrepreneurial education today for future strangers of students" a review of 45 studies conducted in nine countries revealed that despite the inclusion of entrepreneurial learning opportunities in school curricula "students rarely experience this type of educational experience" (Hardie *et al.*, 2020. p.8). Thus, EpE demonstrates its value in promoting entrepreneurial intention and in the development of fundamental skills for acting in a dynamic and uncertain market.

Din *et al.*, (2016), when analyzing university students in Malaysia, showed that entrepreneurial educational approaches contribute to the strengthening of students' self-efficacy, encouraging strategic thinking aimed at risk management and decision making. This finding is in line with the results obtained by Rauch *et al.*, (2015) in the Netherlands, when pointing out that participation in PE programs is associated with increased entrepreneurial attitudes, perceived behavioral control and strengthening the intention to undertake.

The literature (Lanero *et al*., 2011; Nabi *et al*., 2017; Urbancikova; Umarkhonov, 2024; Vecchi *et al*.; 2025) corroborates the relationship between Entrepreneurship Education (EpE) and the development of practical skills. Researches in different countries indicate that EoE boosts management skills, autonomy, innovation, creativity and critical thinking, as well as increasing the entrepreneurial intention of students through immersive and contextualized experiences. Thus, the EoE aims to develop fundamental skills for acting in dynamic, complex and challenging contexts.

2.3 INTEGRATION OF ESG INTO ENTREPRENEURSHIP EDUCATION

The integration of *Environmental, Social and Governance* (ESG) principles to entrepreneurship education in higher education is a strategic axis for the training of professionals capable of responding to complex contemporary challenges. The concept of ESG, consolidated in the business environment as a reference for responsible investments, has been appropriated by education as a normative and pedagogical reference. In this scenario, Entrepreneurship Education (EeE) plays a central role by articulating technical and behavioral skills to sustainability, social equity and ethical governance (Guerrero *et al*., 2020; Cunha *et al*.; 2024).

The environmental dimension of ESG in entrepreneurship education involves the development of critical awareness about the ecological impact of productive and consumption practices. The use of active methodologies, such as Problem Based Learning (PBL), enables students to face real or simulated environmental dilemmas, such as solid waste management, the use of renewable energies and the implementation of the circular economy. This approach favors situated learning and expands the capacity for creative problem solving, forming entrepreneurs able to align competitiveness and sustainability (Hamdan, 2020).

The social dimension of ESG refers to the commitment to diversity, inclusion and collective well-being, which are fundamental for the generation of shared value. In this context, gamified pedagogical practices can create immersive learning environments in which students are exposed to dilemmas related to gender equity, social justice or community impact. By experiencing these scenarios, students internalize values of cooperation, empathy and responsibility, broadening the understanding that entrepreneurial activity cannot be dissociated from the human dimension (Urbancikova; Umarkhonov, 2024).

The dimension of governance is associated with ethics, transparency and *accountability* in management processes. When integrated to entrepreneurship education, this dimension promotes the development of skills related to responsible decision making, participatory leadership and commitment to organizational integrity. Recent studies indicate that entrepreneurs trained in educational environments that value governance are more likely to consolidate business practices based on compliance and social legitimacy (Passarelli; Bongiorno, 2025).

The international literature has emphasized that the incorporation of ESG broadens the perspective of EoE, shifting it from the restricted focus on business creation to an approach oriented to social and institutional transformation. By integrating environmental, social and governance dimensions, entrepreneurial training promotes the development of transversal skills such as critical thinking, empathy and collaborative leadership (Lackéus, 2015; Kirkley, 2017; Cunha *et al*.; 2024). This conceptual widening responds to the need to prepare professionals to deal with the uncertainty and complexity of contemporary ecosystems, where innovation cannot be dissociated from socio-environmental responsibility (Hardie *et al*.; 2020).

Practical experiences indicate that the combination of active methodologies with ESG principles strengthens the entrepreneurial learning process. Board games, simulated juries and case studies based on real problems have been shown to be effective in stimulating ethical reflection and student protagonism. By articulating management challenges with social and environmental values, these pedagogical practices create conditions for the *student to*

understand the interdependence between business success and social commitment (Sailer et al., 2017; Ciprian, 2021).

Despite its potential, ESG integration faces structural and cultural obstacles. The lack of technological infrastructure, the lack of teacher training and curricular rigidity make it difficult to consolidate innovative pedagogical practices in many institutions. In addition, there is a risk that ESG will be reduced to a rhetorical discourse, emptying its transformative character. The literature points out that, to overcome these barriers, it is necessary to strengthen pedagogical intentionality and invest in institutional policies that promote didactic innovation and teacher updating (Børte *et al.;* 2023; Araujo *et al*., 2024).

In this context, the educational entrepreneur (*edupreneur*) emerges as a strategic subject, acting as a mediator between curriculum, institutional innovation and social demands. By articulating active methodologies with ESG values, *edupreneur* extends the traditional function of the teacher, promoting formative experiences that integrate technology, ethics and sustainability. Its performance, marked by political sensitivity and critical awareness, reinforces the role of the university as a space for building transformative entrepreneurial skills (Guerrero; Urbano, 2012; Cui; Bell, 2022).

The integration of ESG to entrepreneurship education should not be perceived as a complement, but as a structuring axis of the training process. By promoting an education that articulates active methodologies and values of sustainability, inclusion and governance, higher education takes a leading role in the preparation of critical and conscious professionals. This perspective responds to the global agendas of sustainable development, reinforcing the role of the university as an agent of innovation and social transformation (UNESCO, 2024).

Finally, consolidating the ESG agenda in entrepreneurship education means training not just business entrepreneurs, but entrepreneurs prepared for the new challenges of the future. These subjects, by combining innovation with ethics and sustainability, play a central role in building more resilient, inclusive and environmentally responsible societies. It is a competitive and civilizing differential, capable of redefining the social function of entrepreneurial education in the 21st century (Nabi *et al*., 2017; Cunha *et al*.; 2024).

3 METHODOLOGY

This study assumes a mixed methodological approach, with a qualitative and quantitative approach, as proposed by Creswell and Clark Plan (2017). This choice is justified by the need to understand how the active Problem Based Learning (PBL) and gamification

methodologies have been applied in the context of PE in higher education. The research is descriptive (Gil, 2008), since it seeks to map perceptions, experiences and practices associated with pedagogical innovation in undergraduate courses and to identify factors that enhance or limit the adoption of these methodologies in HEIs.

The collection of quantitative data was carried out through a questionnaire with closed questions and Likert scale, answered by 66 students of the Administration course of a private university in the state of Pernambuco, between June 27 and 22025. The instrument, elaborated from theoretical references on active methodologies, entrepreneurial skills and ESG principles, aimed to identify perceptions about the use of PBL, gamification and integration of environmental, social and governance values in pedagogical practices. The selection of participants was given by opportunity, considering regular classes in the current academic period (Fachin, 2003).

Three semi-structured interviews were also conducted with teachers and professionals in the area of Entrepreneurial Education, all with recognized performance. The interviews, with an average duration of 45 minutes, were audio recorded with the consent of the participants and later transcribed for the purpose of qualitative analysis. The quantitative data were organized in electronic spreadsheets and treated with the aid of descriptive statistics (frequencies, averages and percentages), allowing to observe patterns and trends in students' responses. The interviews were analyzed using the content analysis technique, according to the systematic procedures proposed by Bardin (2011), aiming at the categorization and interpretation of the speeches in the light of the theoretical framework of the study.

3.1 ANALYSIS AND INTERPRETATION OF DATA

The integration between quantitative and qualitative data allowed to explore the perceptions and experiences related to the use of active methodologies, digital technologies and ESG principles in entrepreneurship education, in addition to capturing nuances about the development of entrepreneurial skills in the context of higher education. Of the 66 participants, with an average age of 29.5 years, 72.7% work full time in the labor market, indicating a profile that values the theory-practice integration. As for the entrepreneurial experience, only 12.1% are active entrepreneurs, while 51.5% have never undertaken, and another 12.1% manifest future intention. This scenario suggests that most students seek in academic training tools applicable directly to their professional reality and entrepreneurial aspirations.

More than half of the students (54.5%) demonstrate intention to undertake, although there is uncertainty in 30.3%. Entrepreneurial self-perception is moderately positive (mean of 3.73), suggesting the need for strategies that reinforce self-confidence. Technology is seen as a strong motivator for entrepreneurship (average of 4.52), emphasizing the importance of integrating digital skills into the curriculum to align training with market reality.

Regarding the relevance of the content of the Entrepreneurship discipline for the professional objectives of the students, the average was 4.06. The data show that most recognize the importance of this training for their career aspirations, although in slightly lower intensity than that related to technology. This reinforces the connection between entrepreneurial education and the life projects of students.

The question "Entrepreneurship education helps solve problems in an innovative way" received an average of 4.15. The statement "The implementation of practical projects, case studies, business simulations and activities that stimulate creativity and problem solving" obtained the highest average among all the questions of the instrument: 4.61. These results indicate recognition of the effectiveness of active methodologies in entrepreneurial training and foster creative problem solving skills.

The use of games, challenges and case studies was well evaluated, with an average of 4.39. Among the respondents, 63.6% gave a maximum grade (5) and 18.2% a grade 4, totaling 81.8% of positive evaluations. These data show the acceptance of active methodologies as tools with this strategy for the teaching of entrepreneurship in higher education.

Regarding the statement "Entrepreneurship education influences my desire to start a business", the average was 3.97, this suggests that education alone is not the only determinant of the decision to undertake. Contextual elements, personal experiences and individual characteristics also seem to influence this motivation, which reinforces the importance of an entrepreneurial training articulated with aspects of the students' reality. Themes such as sustainability, ethics and social inclusion, when inserted in entrepreneurship classes, were evaluated with an average of 3.88. Although lower than the others, this average indicates a positive trend of engagement with content that incorporates social and environmental dimensions into teaching. This indicates the potential of pedagogical practices aligned with ESG principles as a formative differential in the field of entrepreneurship.

Still on the connection between teaching and practice, the statement "The activities of the entrepreneurship discipline contribute to the practical application of knowledge outside the classroom" obtained an average of 4.03, with 66.7% of students assigning grades 4 or 5. These

results highlight the valorization of pedagogical proposals that promote contextualized and applicable learning, strengthening the articulation between theory, practice and reality.

The statement "Practical and interactive methodologies contribute to the development of sustainable solutions aimed at solving real problems" averaged 4.15, with 75.8% of students assigning grades 4 or 5. This result reflects the recognition of students as to the ability of these approaches to promote critical thinking, sustainability and social innovation, pillars increasingly valued in the formation of conscious entrepreneurs who respect the community and differences. The prevalence of students inserted in the labor market, combined with the diversity of entrepreneurial experiences and intentions, reinforces the relevance of dynamic, contextualized and oriented pedagogical approaches to the development of experimental practices.

Three semi-structured interviews were also carried out with teachers and professionals working in the area of Entrepreneurial Education, to which we refer as Respondent 1, Respondent 2 and Respondent 3. Respondent 1 is male, is 39 years old, has a bachelor's degree and a master's degree in Business Administration, and has been teaching in higher education for nine years. He has experience in coordinating projects aimed at university entrepreneurship, with an emphasis on the integration of sustainable practices and social innovation.

Respondent 2 is a woman, is 41 years old, has a degree in Accounting Sciences, a master's and doctoral student in Education and teaches the disciplines of Environmental Management, People Management and Project Management, working in higher education for eight years, develops active methodologies oriented to solving real problems. Respondent 3, on the other hand, is a man, 52 years old, graduated and has a master's degree in Economics and a doctorate in Statistics. With 12 years of teaching experience in higher education, it acts mainly in the training of teachers for the use of new methodologies. Question 3 of the research instrument investigated which entrepreneurial skills are considered relevant, in the perception of experts, for student training in higher education.

> Awakening the entrepreneurial spirit in students is not only teaching how to set up a business. It goes far beyond that. We need to encourage creativity when it comes to solving problems and the ability to act even when things are uncertain. For me, the most important skill today is proactivity, coupled with the ability to work in teams. When the student learns to undertake with purpose, with sensitivity to social and environmental issues, he tends to become a more committed entrepreneur, more aware of the impact that can generate what will help him and the community also in which he is inserted [Respondent 1].

Respondent 1's speech highlights an expanded conception of entrepreneurship, which goes beyond the logic of business creation and incorporates social, ethical and collaborative dimensions. By emphasizing skills such as proactivity, creativity in problem solving and teamwork, the interviewee reinforces the importance of entrepreneurial training focused on situated action and collective responsibility. This perspective is in line with authors such as Lackéus (2015) and Hamdan (2020), who argue that PE should develop transversal skills and transformative attitudes, preparing students to intervene critically in their contexts.

The link between entrepreneurship and social and environmental sensitivity approximates the paradigm of entrepreneurship with purpose, which seeks to align innovation and socio-environmental impact (Kirkley, 2017), while reaffirming the role of the university as an ethical and citizen training space. In this sense, it is observed how the entrepreneurial intentionality, when mediated by collective values, contributes to the constitution of conscious subjects and able to act with responsibility and social commitment. The fifth question of the instrument sought to understand if the respondent has already worked with the ESG principles (Environmental, Social and Governance) in higher education and, if so, how he perceives the integration of these guidelines in the classroom, especially when associated with active methodologies.

> Yes, I work on ESG principles both in the Project Management discipline and in integrative projects in another institution. I see that they create a very rich bridge between the technical content and the more human formation of the students. When I apply methodologies such as Problem-Based Learning, gamification, case studies and even simulated juries, I realize that students begin to reflect more deeply on the impact of the decisions they make not only on the environment but also on the people around them. These methodologies are very effective because they put the student as the protagonist, with ethical responsibility in the learning process. In addition, I realize that they help the student stay. [Respondent 2].

The report of Respondent 2 corroborates the studies of Costa *et al*., (2021) and Malta *et al*., (2025), by highlighting that the integration between active methodologies and ESG principles enhances entrepreneurial learning by articulating technical, ethical and socio-environmental skills. The application of practices such as Problem-Based Learning, gamification and case studies highlights the central role of student protagonism and situated learning, as defended by Lackéus (2015) and Bacich & Moran, (2017), favoring students' cognitive, affective and behavioral engagement, especially in contexts marked by demotivation and evasion, as discussed by Bortolaso *et al*., (2024).

The seventh question was addressed to Respondent 2 and investigated whether he has already applied gamification resources in higher education, requesting that he report some practical experience and their respective results.

> I once developed a board game adapted for the discipline of Environmental Management, in which students needed to make decisions based on ethical and operational dilemmas. The activity was a success because it stimulated both strategic thinking and empathy and dialog among students. Gamification, when well planned, promotes deep engagement and generates rich and memorable debates. However, I realize that if the activity is not well structured or is carried out in a hurried way, the effect can be the opposite, leading students to lose focus and reducing the effectiveness of the proposal. Therefore, whenever I apply a gamified activity, either through board games or using digital platforms such as Kahoot!, Quizlet, Socrative or Nearpod [Respondent 2].

The eighth question sought to understand whether, in the respondent's perception, it is possible to integrate values such as sustainability, inclusion and ethics through educational games, in addition to investigating whether he has already experienced any practical experience in this sense, according to the following report:

> "Yes, it is possible to work on sustainability, inclusion and ethics through games. I have already proposed a project in which students created solutions to simulated environmental problems in an augmented reality game. [Respondent 2]".

Thus, the experience reported by Respondent 2 illustrates the potential of educational games as a formative strategy to integrate the values of sustainability, inclusion and ethics, aligning themselves with the proposals of Urbancikova; Umarkhonov, (2024), who defend gamification as a critical mediation tool in teaching (Lackéus , 2015) and Bacich & Moran, (2017).

The sixth question sought to identify the main challenges faced by teachers in the use of active digital methodologies and the integration of ESG principles (Environmental, Social and Governance) in daily academic life, considering both the structural and formative barriers present in higher education institutions.

> The biggest challenge I realize is related to the lack of adequate equipment and also the lack of teacher training. Many teachers still do not feel comfortable experimenting with active methodologies, especially those that involve the use of technology resources. In addition, we face structural obstacles, such as rigid workloads and the absence of technological support. Overcoming these barriers requires a deeper change in educational institutions. Despite realizing that there is a desire to innovate, there is

> something that curbs this process. I believe that an important step would be to reserve quality time, dedicated exclusively to planning these changes [Respondent 3].

The report highlights structural and formative barriers that limit the adoption of active digital methodologies and the integration of ESG principles in higher education. The lack of infrastructure, teacher training and institutional support aligns with the findings of Costa *et al*., (2021) and Araujo *et al*., (2024), which point to the need for an educational ecosystem favorable to innovation. The emphasis on teacher resistance and work overload reinforces the cultural and operational challenges faced by institutions and requires a planned investment in time, quality and objective conditions for effective pedagogical transformation (Bacich; Moran, 2017). The fourth question of the instrument investigated how entrepreneurial learning can contribute to a more critical formation and aligned with contemporary challenges, considering its transformative potential when articulating theory, practice and social responsibility in the context of higher education.

> Entrepreneurial learning, when articulated with active methodologies and aligned with the student's life context, acts as a true catalyst for transformation. It goes beyond technical content and encourages students to think critically about the world around them, to propose viable and sustainable solutions and to act ethically and responsibly. More than training entrepreneurs in the traditional sense, this approach contributes to the formation of subjects who are able to intervene in reality with creativity and awareness [Respondent 1].

Respondent 1 validates entrepreneurial learning, via active methodologies, as a critical training strategy and student protagonism. Quantitative and qualitative data show that entrepreneurial training in higher education, integrated with active methodologies, digital technologies and ESG principles, develops critical, collaborative and innovative skills. This reinforces the relevance of an expanded education, which articulates theory and practice to form engaged and creative subjects in solving social challenges.

4 FINAL CONSIDERATIONS

This study analyzed how active methodologies, such as PBL and gamification, contribute to PE in higher education. Based on data from students and experts, it was found that these practices favor an ethical, creative and sustainable formation. Games, case studies and projects have been recognized for promoting engagement, critical thinking and innovation, highlighting the transformative potential of interactive approaches to current educational

challenges. The inclusion of ESG principles in pedagogical practices proved to be a relevant differential. The data showed that, when incorporated into entrepreneurship teaching, the environmental, social and governance dimensions broaden the students' training horizon, promoting ethical reflections and awareness of the social responsibilities of the entrepreneur. At the same time, expert testimonials showed that integrating these guidelines requires planning, intentionality, and institutional commitment.

Despite the potential of active methodologies, obstacles persist such as lack of technological infrastructure, insufficient teacher training and curricular rigidity, which hinder their implementation and the integration of ESG-aligned proposals. Even with the willingness of teachers to innovate, the lack of time for planning and continuing education compromises the sustainability of changes. The study highlights the educational entrepreneur as a crucial mediator, integrating didactic, technological and ethical skills to boost an education aligned with contemporary challenges. There is some convergence between the perceptions of students and experts on the benefits of active methodologies and the ESG approach in higher education. This reinforces the need for pedagogical strategies adapted to the various student profiles, considering their experiences and aspirations.

Abstract: Entrepreneurship education in higher education requires a repositioning of teaching practices and institutional structures. Initiatives that incorporate active methodologies, sustainable values and student protagonism demonstrate greater potential to contribute to the formation of critical, innovative subjects committed to social transformation. The adoption of these strategies should not be seen as a passing trend, but as a perennial pedagogical strategy.

Finally, this study contributes to the debate on pedagogical innovation in higher education, by highlighting the importance of integrating the formative intentionality to entrepreneurial educational practices. It is recommended, for future research, the deepening of the analysis on the role of the educational entrepreneur and the impact of institutional policies to support didactic innovation. It is also relevant to explore interdisciplinary and collaborative experiences that strengthen an educational culture oriented towards social transformation.

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